

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Casio
Firmware Version: N/A
Model(s): XJ-F210WN, XJ-F200WN, XJ-F20XN, XJ-F100W, XJ-F10X, XJ-V10X, XJ-V100W, XJ-V110W

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0001-b003	1.4.1

Version History

Module Version	Date	Notes
1_2_0_0	10/31/2017	Added ethernet support for models: XJ-F210WN, XJ-F200WN, XJ-F20XN. Removed unsupported Video Mute command from Ethernet control. Added Audio Mute and AV Mute commands for Ethernet control.
1_1_0_0	12/14/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'admin'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='XJ-F210WN')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='XJ-F210WN')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='XJ-F210WN')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ⁹	Value '16:9' 'True' ^{2,6} 'Normal A' ²	Value 'Letter Box' ^{4,6} '4:3' ⁸ 'Normal B' ³	Value 'Full' ² '16:10'
# AspectRatio example InterfaceName.Set('AspectRatio', '16:9')			
Command AudioMute ¹⁰	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AVMute ¹⁰	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command Freeze ⁹	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input ⁹	Value 'RGB' 'HDMI 1' ⁴ 'USB Display' ⁴ 'Auto' ⁹	Value 'Component' 'HDMI 2' ⁴ 'S-Video' ⁴ 'CASIO USB Tool' ^{4,9}	Value 'Video' ⁴ 'Network' ⁴ 'HDMI' ⁵ 'File Viewer' ^{4,9}
# Input example InterfaceName.Set('Input', 'RGB')			
Command MenuNavigation ⁹	Value 'Up' 'Right' 'Menu'	Value 'Down' 'Escape'	Value 'Left' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command MicInputLevel ^{1,9}	Value '0' – '3'		

# MicInputLevel example InterfaceName.Set('MicInputLevel', '0')			
Command Mute ⁹	Value 'On'	Value 'Off'	
# Mute example InterfaceName.Set('Mute', 'On')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Transport ⁹	Value 'Play/Pause' 'Previous'	Value 'Rewind' 'Next'	Value 'Fast Forward'
# Transport example InterfaceName.Set('Transport', 'Play/Pause')			
Command VideoMute ⁹	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume ⁹	Value 0 to 30 in steps of 1		
# Volume example InterfaceName.Set('Volume', 30)			

¹ Command only available on models XJ-F210WN, XJ-F200WN and XJ-F20XN

² Only available when input is set to RGB or a PC HDMI source

³ Only available when input is set to Component, Video, S-Video or a DTV HDMI source

⁴ Not available on XJ-V10X, XJ-V100W, XJ-V110W

⁵ Only available on XJ-V10X, XJ-V100W, XJ-V110W

⁶ Not available on XJ-V10X, XJ-F10X, XJ-F20XN

⁷ Not available on the XJ-V100W, XJ-V110W, XJ-F100W, XJ-F200WN, XJ-F210WN when it is set to RGB or a PC HDMI source

⁸ Not available on the XJ-V10X, XJ-F10X, XJ-F20XN when it is set to RGB or a PC HDMI source

⁹ Not available for Ethernet control.

¹⁰ Not available for Serial control.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio ⁹	Value '16:9' 'True' ^{2,6} 'Normal A' ²	Value 'Letter Box' ^{4,6} '4:3' ⁸ 'Normal B' ³	Value 'Full' ² '16:10'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute ¹⁰	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command AVMute ¹⁰	Value 'On'	Value 'Off'	
# AVMute examples InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Normal' 'Lamp Warning' ¹⁰	Value 'Fan Warning' ¹⁰ 'Lamp Error'	Value 'Fan Error' 'Temperature Warning' ¹⁰

Global Scripter Module Communication Sheet

Revision: 10/31/2017

	'Temperature Error'	'Cover Warning' ¹⁰	'Cover Error'
	'Filter Warning' ¹⁰	'Filter Error'	'Other Warning' ¹⁰
	'Other Error'	'Multiple Warnings / Errors'	'Light Error'
# DeviceStatus examples InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command Freeze ⁹	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'RGB' 'HDMI 1' ⁴ 'USB Display' ⁴ 'Auto' ⁹	Value 'Component' 'HDMI 2' ⁴ 'S-Video' ⁴ 'CASIO USB Tool' ^{4,9}	Value 'Video' ⁴ 'Network' ⁴ 'HDMI' ⁵ 'File Viewer' ^{4,9}
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampHours	Value Hours		
# LampHours examples InterfaceName.Update('LampHours') Value = InterfaceName.ReadStatus('LampHours') InterfaceName.SubscribeStatus('LampHours', None, FeedbackHandler)			
Command MicInputLevel ^{1,9}	Value '0' – '3'		
# MicInputLevel examples InterfaceName.Update('MicInputLevel') Value = InterfaceName.ReadStatus('MicInputLevel') InterfaceName.SubscribeStatus('MicInputLevel', None, FeedbackHandler)			
Command Mute ⁹	Value 'On'	Value 'Off'	
# Mute examples InterfaceName.Update('Mute') Value = InterfaceName.ReadStatus('Mute') InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)			
Command Power	Value 'On' 'Warming Up'	Value 'Off'	Value 'Cooling Down'

<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		
Command VideoMute ⁹	Value 'On'	Value 'Off'
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>		
Command Volume ⁹	Value 0 to 30 in steps of 1	
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>		

¹ Command only available on models XJ-F210WN, XJ-F200WN and XJ-F20XN

² Only available when input is set to RGB or a PC HDMI source

³ Only available when input is set to Component, Video, S-Video or a DTV HDMI source

⁴ Not available on XJ-V10X, XJ-V100W, XJ-V110W

⁵ Only available on XJ-V10X, XJ-V100W, XJ-V110W

⁶ Not available on XJ-V10X, XJ-F10X, XJ-F20XN

⁷ Not available on the XJ-V100W, XJ-V110W, XJ-F100W, XJ-F200WN, XJ-F210WN when it is set to RGB or a PC HDMI source

⁸ Not available on the XJ-V10X, XJ-F10X, XJ-F20XN when it is set to RGB or a PC HDMI source

⁹ Not available for Ethernet control.

¹⁰ Not available for Serial control.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

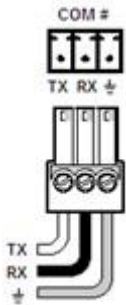
Data Bits: 8

Parity: None

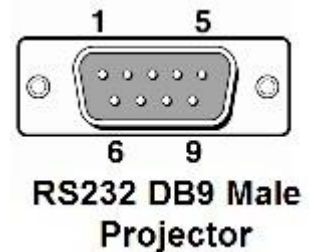
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	4352
Logon Credentials Supported:	Yes
Default Password:	admin
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands (Serial)

Aspect Ratio 16:10	(ARZ7)
Aspect Ratio 16:9	(ARZ1)
Aspect Ratio 4:3	(ARZ6)
Aspect Ratio Full	(ARZ4)
Aspect Ratio Letter Box	(ARZ3)
Aspect Ratio Normal A	(ARZ0)
Aspect Ratio Normal B	(ARZ2)
Aspect Ratio True	(ARZ5)
Freeze Off	(FRZ0)
Freeze On	(FRZ1)
Input Auto	(SRC6)
Input CASIO USB Tool	(SRC13)
Input Component	(SRC1)
Input File Viewer	(SRC11)
Input HDMI	(SRC7)
Input HDMI 1	(SRC7)
Input HDMI 2	(SRC14)
Input Network	(SRC8)
Input RGB	(SRC0)
Input S-Video	(SRC9)
Input USB Display	(SRC12)
Input Video	(SRC2)
Menu Navigation Down	(KEY2)
Menu Navigation Enter	(KEY5)
Menu Navigation Escape	(KEY6)
Menu Navigation Left	(KEY3)
Menu Navigation Menu	(KEY11)
Menu Navigation Right	(KEY4)
Menu Navigation Up	(KEY1)
Mic Input Level 0	(MIC0)
Mic Input Level 3	(MIC3)
Mute Off	(MUT0)
Mute On	(MUT1)
Power Off	(PWR0)
Power On	(PWR1)
Transport Fast Forward	(KEY28)
Transport Next	(KEY30)
Transport Play/Pause	(KEY26)
Transport Previous	(KEY29)
Transport Rewind	(KEY27)
Video Mute Off	(BLK0)
Video Mute On	(BLK1)

Volume 0	(VOL0)
Volume 30	(VOL30)

Appendix B. Query Commands (Serial)

Aspect Ratio	(ARZ?)
Device Status	(STS?)
Freeze	(FRZ?)
Input	(SRC?)
Lamp Hours	(LMP?)
Mic Input Level	(MIC?)
Mute	(MUT?)
Power	(PWR?)
Video Mute	(BLK?)
Volume	(VOL?)

Appendix C. Set Commands (Ethernet)

Audio Mute Off	%1AVMT 20\x0D
Audio Mute On	%1AVMT 21\x0D
AV Mute Off	%1AVMT 30\x0D
AV Mute On	%1AVMT 31\x0D
Input Component	%1INPT 23\x0D
Input HDMI 1	%1INPT 31\x0D
Input HDMI 2	%1INPT 32\x0D
Input Network	%1INPT 51\x0D
Input RGB	%1INPT 11\x0D
Input S-Video	%1INPT 22\x0D
Input USB Display	%1INPT 52\x0D
Input Video	%1INPT 21\x0D
Power Off	%1POWR 0\x0D
Power On	%1POWR 1\x0D

Appendix D. Query Commands (Ethernet)

AV Mute	%1AVMT ?\x0D
Device Status	%1ERST ?\x0D
Input	%1INPT ?\x0D
Lamp Hours	%1LAMP ?\x0D
Power	%1POWR ?\x0D